somewhat in the manner of inlaid work. By what the Dutch could understand, one of these idols was called *Taurico*, and the other *Dago*, as by these words they address them. They paid great regard to these idols, clapping their hands and jumping, when they approached them; and being in great terror when the great guns were fired among them, they applied, as it should seem, to one of their gods for assistance, shouting and howling out *Dago*! *Dago*! whose succour they thought so necessary in that time of danger.

After having experienced a great storm while they lay before Easter Island, they departed from thence on the 12th April; ...."

Conclusions

Our conclusions, after studying the aforementioned account of Jacob Roggeveen's voyage, are the following:

a) In 1722, the men in Easter Island were tall and had heavy-boned and well proportioned bodies. They painted themselves with red and dark brown colours. The women were not as tall as the men and they were painted with a scarlet colour.

b) The Dutch observed that the *pukáo* "...was adorned with a garland". Thirty-eight years after, in 1770, the Spanish sailors drew on the maps of the island some *moái* with their *pukáo* and marked the crowns with little dots. This explains that some of the "cupules pecked" found in several *pukáo* correspond to the crowns mentioned by the Dutch and drawn by the Spaniards. Consequently, the *pukáo* had these cupules when they were placed over the *moái's* heads. The cupules very probably had white coral incrustations as well as the eyes did, or maybe they had just white stones showing the difference with the red colour of the *pukáo*.

In our opinion, since the *moái* represented a deceased person, they were possibly painted with the same or similiar colours and had the same or similar drawings and ornaments as the deceased, including the hat or crown.

Thomson (1891), Lavachery (1939), Mellén (1986 and 1990), Van Tilburg and Lee (1987), Van Tilburg (1992) and Lee (1992) made refernces to the hats and drawings of the *moái* as well as to the pecked cupules of the *pukáo*.

c) The first name of a *moái* that we know of are *Taurico* (maybe *Tau riku*, *Ta uri ko* or *Tuu riku*, and *Dago* (*Ta kou*). These names must be accepted with the maximum linguistic precaution.

d) John Hamilton Moore's account includes several errors, i.e., the native's height of 12 feet (3.66 meters).

Note

RNJ readers will be able to compare John Hamilton Moore's account with the Roggeveen and Bouman Journals, translated by Mr. von Saher and published by *RNJ* (1990), vol.4, No.3 and (1990-91), vol.4(4).

References

Behrens, Carl F. 1739. *Histoire de l'expédition de trois vaisseaux envoyés par la Compagnie des Indes Occidentales des Provinces-Unies aux Terres Australes en MDCCXXI.* 2 vols. La Haye.

Lavachery, Henri. 1939. *Les pétroglyphes de Ile de Pâques.* 2 vols. De Sikkel, Anvers.

Lee, Georgia. 1992. *Rock Art of Easter Island: Symbols of Power, Prayers to the Gods.* UCLA Institute of Archaeology.

Mellén, Francisco. 1986. *Manuscritos y documentos españoles para la historia de la isla de Pascua.* CEHOPU, Madrid.

_____1990. Cartografía histórica de la isla de Pascua en el siglo XVIII, acompañada de algunos datos etnológicos y arqueológicos. *Courier Forsch. Inst. Senckenberg,* 125:123-137. Frankfurt.

Moore, John Hamilton. 1778. *A New and Complete Collection of Voyages and Travels...* London.

Thomson, William J. 1891. Te pito te Henua, or Easter Island. In *Report of the U.S. National Museum for the year ending 30 June, 1889.* Washington D.C.

Van Tilburg, Jo Anne. 1992. *HMS Topaze* on Easter Island: Hoa Hakananai'a and Five other Museum Sculptures in Archaeological Context. British Museum *Occasional Papers 73,* Department of Ethnography.

Van Tilburg, J. and Lee, G. 1987. Symbolic Stratigraphy: Rock art and the monolithic statues of Easter Island. *World Archaeology,* 19(2):133-45.

Dr. Paul Bahn reports that AMS Press has located a box of Routledge's book, *The Mystery of Easter Island*, originally published in 1919 and reprinted by AMS in 1978. It was formerly thought that all copies had been sold. Those interested in obtaining a copy of this reprint, contact AMS Press, New York, NY. The ISBN number is 0-404-14231-1.

LETTERS TO THE EDITOR

✉ The following article is a response to Christian Walter, *RNJ* 7(2):32:

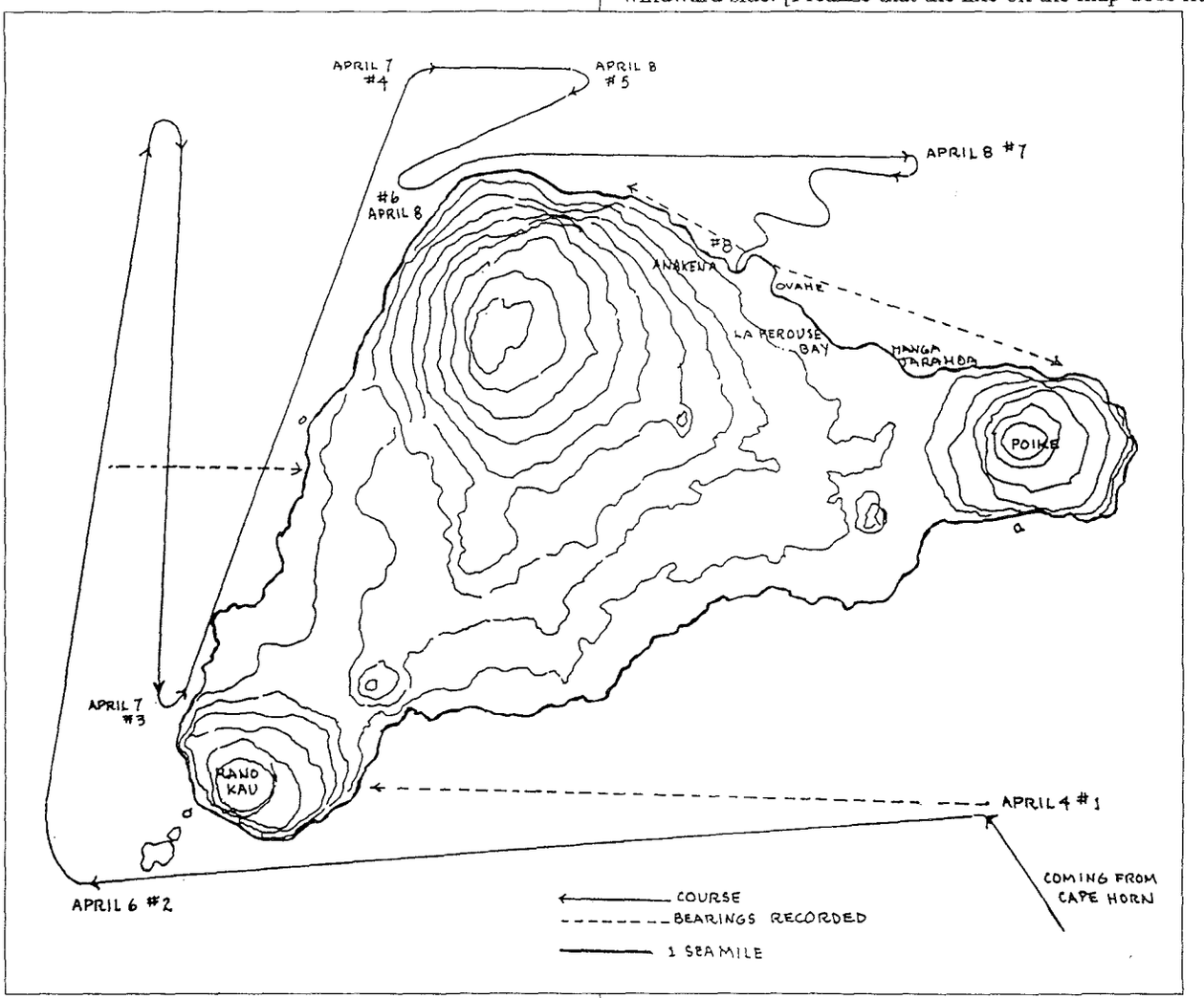
MORE DETAILS FROM THE JOURNALS OF JACOB ROGGEVEEN AND CORNELIS BOUMAN, NAUTICAL DEPARTMENT

Herbert von Saher, The Netherlands

When I wrote my articles under the title of "Some Details from the Journal of...." the discoverers of Easter Island (RNJ, Volume 4, Nos. 3 and 4) I thought that the readership of RNJ was more interested in information in the field of archaeology and ethnography than in navigation and meteorology. The journals of the old seafarers consist of a large part of this nautical information, for it forms the justification to those that ordered and financed their voyage, it shows what was possible and impossible under the given circumstances and it serves future navigators to find their routes.

So I left out a lot, thinking that RNJ readers would not be interested and would find these comments on navigation boring. But now I have been challenged by a reader who is of

the opinion that even my limited indications in this field contained mistakes and that I got Roggeveen's landing site on Easter Island wrong; moreover, the editors of RNJ have been criticized for printing my mistakes without properly checking first.



Map showing the course and bearings taken by Roggeveen as he circled around the island.

I must defend myself and I do this by producing a map upon which I indicated the course that the Dutch fleet followed in those first days of April 1722, the bearings they took on the coast and the site where they finally anchored and landed. With every change of course recorded in the journals, there is a numbered note on the map (#1, etc.) that corresponds with the notes given below. Now we take the journals of Roggeveen and Bouman in hand and follow them from day to day. I follow Roggeveen's text and only indicate when Bouman is quoted; he has fewer entries, but is sometimes more exact. [The translation from the Linschoten edition is my own.]

Sunday 5 April (Easter Day), #1: Late in the afternoon the *Africaansche Galey* [the smallest ship in the fleet of three] sighted land West by North at 5½ sea miles. It was decided to lower sails and stand off during the night until daylight. Roggeveen decides to call the island "Easter Island".

Monday 6 April: In the morning they still have the island in the West at 8 to 9 miles, wind East South East. They changed course (#2) from West to North West in order to reconnoiter the lee side of the island and not run the risk of being on the windward side. [I realize that the line on the map does not run WNW, but if it did, it would leave the map.]

When they have the island at 2 miles in the East the captains of the three ships come together in 'ship's council'. From the smoke they have seen they conclude that the island is inhabited; they want to orient themselves and they plan a landing party for the next day, hoping to find a suitable anchorage and to get some vegetables, fresh fruit, and meat. With that purpose in mind they keep their ships near to the coast during the night (but without dropping anchor because the water was too deep). They must have been moving to and fro in the lee side of the island.

Tuesday 7 April (#3): Very unstable weather, thunder, lightning, heavy showers, changing winds from the North West, with the result that the West coast had now become the windward side. Under those weather conditions it would be impossible to land. Bouman: we kept moving near the coast. This was the day the naked fisherman from Rapa Nui came on board [see page 50, RNJ 4(4)]. Bouman: at 20.00 hours we changed course to East, (#4) strong winds during the night.

Wednesday 8 April, Bouman: at 2.00 hours we changed course to South West (#5).

Bouman: at daylight we came very near the Northwest corner of the island and then changed course to East (#6); we saw some heathen statues on land and many people.

Roggeveen: after breakfast we sent two sloops to the coast for orientation but with the instruction not to land if there were too many people on the beach [because of the danger of a small landing party being attacked by a large group of islanders].

The sloops returned without landing because there were too many people about. They reported that the islanders --unlike the first fisherman seen--were "smartly" dressed and made friendly signs inviting the sailors to come ashore.

Bouman: we started tacking towards the coast (#7) and at 11.00 hours we were so near the coast that we could drop anchor in 23 fathoms. The soil was grayish white coral-like sand.

Roggeveen: at sunset we arrived on the roads between the two other ships that had already anchored there; we dropped anchor at 22 fathoms, coral ground, a quarter mile from land.

So this was the first time the fleet dropped anchor at Easter Island after slowly moving around for three days in a clockwise direction. [The accusation that I miscalculated the landing place once, on page 34, "...this must have been along the West coast" is beside the point. I stated that the fleet was cruising along the West coast, no one suggested a landing on the 6th of April.]

Where were they now? Both Roggeveen and Bouman gave the5 same orientation:

Roggeveen (#8): we took a bearing on the Eastern corner of the island, we had it East by South and the Western corner West North West of us. Bouman speaks of the "Northern corner" in the East (evidently meaning the Poike peninsula) and the Northwestern corner also at West North West [I tried to explain that already on page 50.] These were of course the Eastern and Northern corners that they could see from their ships, not the farthest points East and North of the island.

From these two bearings it becomes irrefutably clear that they were somewhere on the North coast of the island. Bouman adds another bearing: we have a big, erected statue, standing near the shore South West by West.

We do not know what *moai* were still standing in 1722, but this *moai* standing near the shore limits the landing site to only four possibilities: Anakena, La Perouse Bay, Hanga Tarahoa, and possibly the vicinity of Playa de Ovahe as these are the only places on the North coast where *moai* have been found or could be clearly seen from the sea. Why do swimmers only go to Anakena and Ovahe? Because these are the only places with a nice sandy beach, the grayish white sand that Roggeveen as well as Bouman noticed on their anchors. At La Perouse Bay and Hanga Tarahoa there is only hard rock.

Now we follow the journals again, because there is more relevant information coming.

April 9, Bouman: for the first time nice weather, we used this quiet day to careen our ships in order to clean the bottoms.

Roggeveen: many islanders came on board bringing chickens, bananas, yams for which they did not ask payment, but also stealing hats, etc. The ships remain at anchor.

April 10, Bouman: at 7:00 the landing party goes ashore, he then gives a long description of the island already reproduced on pages 51-52 in Vol. 4(4).

Roggeveen: In the morning the landing party of 134 men (being the greater part of the whole crew) leaves in five sloops [see page 34 in RNJ Vol. 4(4)].

Later in the day when the King of Rapa Nui invites the Dutch party to the other side of the island where they understood that agriculture and fruit trees were to be found, they noticed that a strong Northerly wind was starting so that their anchorage was suddenly on the windward side of the island. This very much concerned the sailors, because a dangerous situation had arisen: they had left too few men on board to be able to handle this emergency. So it was decided to return immediately to the ships. After their return they wanted to sail away.

However, 11 April still finds them at anchor, with a strong wind blowing North-North West to North West. They cannot sail out of the bay with the wind coming from this direction. They even lower the yards from the main and foremasts (in order to catch less wind). The anchor rope of the *Thienhoven* snaps, but its crew can manage the situation.

12 April: a very bad day. As a result of the strong wind in combination with heavy seas, the anchor ropes of two ships snap; now they are in great danger of being thrown on the rocks. They raise the yards again and prepare for a desperate sailing, and then luck is with them: during a shower the wind turns to the West and at last they can sail out of the bay clear of the rocks. At sunset they have the Eastern corner of the island at six miles to the South West and they do not intend to return.

13 April: the fleet turns to the west to continue their journey in search of the Southland.

What do we learn from these events? In a strong Northerly wind these ships are unable to escape from the bay. When we now look at the map, we see that from Playa de Ovahe, from La Perouse Bay and from Hanga Taharoa you can easily sail out to the East at a 90° angle with the Northerly wind. On the 11th, when they would dearly have loved to sail out, for they were two whole days in great danger of being thrown on the rocks, Roggeveen described the wind as NNW and NW. Only in Anakena Bay is one trapped and cannot move out in the North East direction. In my opinion this was the decisive factor that excludes all the alternatives: the high hill called Maunga Puha blocks the way, thus making it too much against the wind for the square-rigged ships of the 18th century. So I maintain my contention as set out in RNJ (1991:50-51) that the landing was at Anakena.

Since reaching this conclusion, I have been trying to find reasoning by others trying to locate Roggeveen's landing site. The really authoritative source, the edition of Roggeveen's journal by the "Linschoten Vereeniging", (the Dutch equivalent of the English Hakluyt Society), generally edited by retired admirals who are very critical of the navigation mistakes they manage to find in the old journals. Roggeveen gets very

high marks for his calculation of the position of Easter Island on the globe, which was only 1°39' wrong (actual position: 109° 20' West).

In those days the exact chronometers needed to calculate longitude were not yet available and it had been months before when they had left their last fixed point of orientation. Roggeveen's own indication of the anchorage has been given above under #8, but the editor of Roggeveen's journal did not try to identify it in present-day terms.

The anonymous account "Tweejarige Reyze rondom de Wereld" of 1728 contains no indication. In Behrens' French edition (1739:125): "We entered to the Southeast into a bay in order to drop anchor there." No further details are given. But Andrew Sharp says (1970:97) that Behrens indicated La Perouse bay. Where he found this, I do not know. [Don't forget that La Perouse arrived only 66 years after Behrens had left.] Anyhow, Behrens was only a soldier, whose knowledge of navigation was nil, as can easily be demonstrated from many items in his report. Behrens gives an indirect reference however, saying later on April 10th that a strong Westerly wind developed, preventing the fleet from leaving the bay where they were at anchor (if this were true it would be even stronger support for the Anakena thesis); whereas Roggeveen himself speaks of a Northerly wind (Mulert 1911:124). Claret Fleurieu (1790:300), evidently basing himself on Behrens' information, concludes erroneously: "Because a Westerly wind obliges Roggeveen to leave his anchoring site, it is probable that he had dropped anchor at the West coast and most likely in the Southern part, it is also in this part that Cook and La Perouse anchored".

When Roggeveen's journal was finally found in 1836, 114 years after it had been written, it was first published by the "Zeeuwsch Genootschap der Wetenschappen" [Scientific Society of Zealand] in Middelburg in 1838. The writer compared the navigational part of Roggeveen's journal with previous reports and in 5 Appendices he gives 15 pages of corrections on the old reports. There is a lot of discussion on Easter Island's position on the globe, but no attention is given to the landing site itself. C.E. Meinicke repeats this exercise in 1874, giving Roggeveen very high marks for his navigation and the exactitude of his methods. But unfortunately he also limits himself to the general location of Easter Island and other islands discovered by Roggeveen and makes no mention of the landing site.

Finally we have the excellent translation of Roggeveen's own journal by Andrew Sharp (1970). He says himself that Roggeveen anchored at La Perouse Bay (pg. 93), maybe also based on the alleged indication by Behrens that I could not locate, but at all events he gives no reasoning why it should have been there. Paul Bahn also writes in *Easter Island Studies* (1993:53) that Roggeveen landed at La Perouse Bay, without mentioning his source, but I suppose that it is Andrew Sharp. Final conclusion: there is no authoritative and reasoned identification of Roggeveen's landing place. In the light of this conclusion, I see no reason to change my thesis of Anakena.

I did make a mistake in my first article on Roggeveen's journal (RNJ 4(3):35) in describing the anchorage as being "on the northwest coast" whereas it was on the North coast. I made this error prior to my investigations for Vol.4(4) and I considered it too slight to correct

I did not add a bibliography to my two articles in 1990 and 1991 because the text clearly stated that I had only one source: the journals of Roggeveen and Bouman (in the 1911 edition of the Linschoten Vereeniging, translated by myself).

.

References

Anonymous. 1727. Kort en Nauwkeurig Verhaal van de Reize door 3 Schepen. In *'t jaar 1721 gedaan* [Two editions, both Amsterdam].

Bahn, Paul G. 1993. The History of Human Settlement on Rapanui. In: *Easter Island Studies*, edited by Steven Roger Fischer. Pp.53-55. Oxford: Oxbow Books.

Behrens, C.F. 1728. *Tweejarige Reyze rondom de Wereld.* Dordrecht.

Claret Fleurieu, C. P. 1798/1800. *Voyage autour du Monde pendant les Annees 1790, 1791 et 1792 par Etienne Marchand, auquel on a joint un examen critique du Voyage de Roggeveen.* Paris.

Corney, B.G. 1908. *The Voyage of Captain Don Felipe Gonzalez...to Easter Island, 1770-1.* Hakluyt Society, Cambridge.

Meinicke, C.E. 1874. Jacob Roggeveen's Erdumseglung in 1721-1722; *Eleventh Annual report of the Verein fuer Erdkunde.* Dresden.

Mulert, F.E. 1911. *De Reis van Mr. Jacob Roggeveen.* Linschoten Vereniging, den Haag.

Roggeveen, Jacob. 1838. *Dagverhaal der Ontdekkingsreis van Mr. Jacob Roggeveeen in de jaren 1721 en 1722.* Zeeuwsch Genootschap der Wetenschappen, Middleburg.

Sharp, Andrew. 1970. *The Journal of Jacob Roggeveen.* Oxford.

von Saher, Herbert. 1990/1. Some Details from the Journal of Captain Bouman on the Discovery of Easter Island. *Rapa Nui Journal,* 4(4):49-52

von Saher, Herbert. 1992. More Journals on Easter Island: The works of Johann Reinhold Forster (1729-1798) and Johann George Adam Forster (1754-1794. *Rapa Nui Journal* 6(1):1-5.

von Saher, Herbert. 1992. More Journals on Easter Island (continued). *Rapa Nui Journal* 6(2):34-39. ❑